

Proven C Book

Modern C, built on the proven library

draft — translation in progress

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An introduction to C, and an introduction to the proven C library.

Written for readers who are just starting out with C.

Translation status

This English edition is being translated from the Korean original, which is complete (12 parts, 58 chapters, appendices A–E and an index, about 570 pages). The translation is **in progress**: this volume currently carries the front matter, and chapters are added as they are translated.

Until then, the complete text is the Korean edition:

- PDF — [dist/proven_c_book-v0.1.0.pdf](#)
- Web — [rubicus-api.github.io/proven_c_book/ko/](#)

Chapter-level progress is tracked in the repository ([TRANSLATION.md](#)).

Preface

This book is an introduction to **proven**, a C library I wrote — and, honestly, a piece of promotion for it.

But to argue that something is worth using, you first have to show why it is needed. The problems **proven** addresses live in the subtle corners of C: the limits of representation, the rules of conversion, the lifetime of memory, the regions the standard declines to define. To someone who has not met those corners, **proven** looks like an unnecessary contraption. So I decided to explain the problems first. That is why most of this book is plain standard C, and why **proven** does not appear in earnest until Part XI. If, after the earlier chapters, you conclude that you do not need **proven** — that is a good outcome too.

There is also, in these pages, some measure of affection for the language. Try explaining why a fifty-year-old language is still here, and what its design gave up in order to get what it got, and affection follows on its own. I have not hidden that part.

Every listing in this book is really compiled and really run. The printed output was not transcribed by hand: it is what the machine produced when the book was built, cross-checked with two compilers (gcc and clang).

One more disclosure. **This book was written with AI as an assisting tool.** What to cover, in what order, under which principles, and what to leave out — all of that I decided, and the draft was written to those instructions and then reviewed by me. Technical claims were checked against the standard and primary sources, and the examples are verified by machine on every build: whoever wrote it, **code that does not run does not go into this book.**

Errors will remain nonetheless. Those are mine, not the tool's. If you find one, please tell me.

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The example code is licensed under the **MIT license**. Anything you learn here can go into your own programs without restriction.

Contact is by email. Corrections are best filed through the GitHub repository — they stay on the record and other readers can see them.

Only error reports are accepted as contributions. Wrong statements, listings that do not behave as printed, typos and stale information are all welcome. Manuscript contributions — new chapters or sections — are not.

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